

Guoqing Chao

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Research Profile

Guoqing Chao is a Professor and PhD Advisor in the School of Computer Science and Technology at Harbin Institute of Technology, Weihai, China. His research interests include **machine learning, data mining**. His **Google scholar citation is 2306, h-index is 21 and i10-index is 31 (Until 2026-9-1)**.

His recent research focuses particularly on **multi-view representation learning, incomplete multi-view learning, deep clustering, multimodal learning, anomaly detection, and intelligent data analysis**. His research has been published in leading journals and conferences in artificial intelligence, machine learning, data mining, and information fusion, including **TPAMI, TKDE, TIP, TMM, TNNLS, TCSVT, IEEE Transactions on Big Data, Information Fusion, Information Sciences, Knowledge-Based Systems, EAAI, ICML, KDD, AAAI, ACM MM, CVPR and IJCAI**. He has extensive experience in **peer review, editorial activities, conference program committees, workshop organization, and special-issue editing**, making him well qualified to serve as an Editorial Board Member or Associate Editor for journals in machine learning, artificial intelligence, data mining, pattern recognition, and intelligent information systems.

Education

Ph.D. in Computer Applied Technology, 2015

East China Normal University, China

Supervisor: Prof. Shiliang Sun

B.S., 2009

Xinyang Normal University, China

Research Experience

Professor / Faculty Member

Harbin Institute of Technology, Weihai, China

2020–Present

Research Scientist

School of Information Systems, Singapore Management University, Singapore

2018–2020

Research Laboratory / LARC

Supervisor: Prof. Ee-Peng Lim

Postdoctoral Fellow

Feinberg School of Medicine, Northwestern University, USA

2017–2018

Supervisor: Prof. Yuan Luo

Postdoctoral Fellow

School of Engineering, University of Connecticut, USA

2015–2017

Supervisor: Prof. Jinbo Bi

Selected Journal Publications

1. [Guoqing Chao*](#), Mingjie Zhang, Xiru Wang, Jie Wen, Weiping Ding, Dianhui Chu. "TrashToTreasure: An Informative and Interactive Multi-View Classification Framework, TKDE, 2026.
2. Shicong Zeng, [Guoqing Chao*](#), Junquan Wei, Yanwei Yu, Zhijin Wang, Dianhui Chu. "LaGraph: Laplacian-Guided Graph Learning for Time Series Anomaly Detection", TKDE, 2026.
3. [Guoqing Chao](#), Jingnan Qi, Jingxuan Li, Dianhui Chu, "Incomplete Multi-view Classification with Heterogeneous Graph", Information Fusion, 121:103137, 2025.
4. Mingyu Wang, Guoqing Chao, Xi Wang, Yongyong Chen, Xijiong Xie, Dianhui Chu. "Multi-modal data augmentation based on masked modeling for image-text retrieval", Knowledge-Based Systems, 113821, 2025.
5. Xuejiao Yu, [Guoqing Chao*](#), Yi Jiang, Jie Wen, Dianhui Chu, "Incomplete Multi-View Clustering via Mutual Information", IEEE Transactions on Multimedia, 2025.
6. Xuejiao Yu+, Yi Jiang+, [Guoqing Chao*](#), Dianhui Chu. "Deep Contrastive Multi-View Subspace Clustering with Representation and Cluster Interactive Learning", IEEE Transactions on Knowledge and Data Engineering, ([ESI Highly Cited Paper](#)) 2024.
7. [Guoqing Chao](#), Kaiwen Zhang, Xiru Wang, Dianhui Chu. "Three-teaching: A three-way decision framework to handle noisy labels", Applied Soft Computing, 111400, 2024.
8. Jianjun Su, Xuejiao Yu, Xiru Wang, [Guoqing Chao*](#), "Enhanced Transfer Learning with Data Augmentation", Engineering Application of Artificial Intelligence, 129:107602, 2024.
9. [Guoqing Chao*](#), Jingyao Liu, Mingyu Wang and Dianhui Chu. "Data Augmentation for Sentiment Classification with Semantic Preservation and Diversity", Knowledge-Based Systems (accepted, 中科院一区TOP期刊), 280:111038, 2023.
10. Guanzhou Ke, [Guoqing Chao](#), Xiaoli Wang, Chenyang Xu, Yongqi Zhu, Yang Yu, "A Clustering-guided Contrastive Fusion for Multi-view Representation Learning", IEEE Transactions on Circuits and Systems for Video Technology, ([ESI Highly Cited Paper](#)), 2023.
11. Zhaoyi Yuan, Hao Ding, [Guoqing Chao*](#), Mingqiang Song, Lei Wang, Weiping Ding and Dianhui Chu. "A Diabetes Prediction System Based on Incomplete Fused

- Data Source", Machine Learning & Knowledge Extraction, 5(2): 384-399, 2023.
12. [Guoqing Chao](#), Songtao Wang, Shiming Yang, Chunshan Li, Dianhui Chu, "Incomplete multi-view clustering by multiple imputation and ensemble clustering". Applied Intelligence (IF=5.086, 52: 14811–14821, 2022). DOI: 10.1007/s10489-021-02978-z.
 13. Chunshan Li, Yingli Zhou, [Guoqing Chao](#) and Dianhui Chu, "Understanding users' requirements precisely: a double Bi-LSTM-CRF joint model for detecting user's intentions and slot tags", Neural Computing and Applications (IF= 5.606, 34:13639–13648, 2022).
 14. Zhijin Wang and Qiankun Sun and [Guoqing Chao](#) and Bing Cai and Yaohui Huang and Yonggang Fu, "A Multi-view Time Series Model for Share Turnover Prediction", Applied Intelligence (IF=5.086, 52:14595–14606, 2022).
 15. Feixiang Sun, Xijiong Xie, Jiangbo Qian, Yu Xin, Chong Wang and [Guoqing Chao](#), "Multi-View K-Proximal Plane Clustering" , Applied Intelligence (IF=5.086, 52:14949–14963, 2022).
 16. Shuqin Wang, Yongyong Chen, Shuang Yi, [Guoqing Chao](#), "Frobenius norm regularized robust graph learning for multi-view subspace clustering". Applied Intelligence (IF=5.086, 52: 14935-14948, 2022).
 17. [Guoqing Chao](#), Shiliang Sun, Jinbo Bi, "A survey on multi-view clustering", IEEE Transactions on Artificial Intelligence (2(2): 146-168, 2021). DOI : 10.1109/TAI.2021.3065894.
 18. [Guoqing Chao](#) and Shiliang Sun, "Semi-supervised multi-view maximum entropy discrimination with expectation Laplacian regularization". Information Fusion (IF = 11.21, pages 296-306, 2019).
 19. [Guoqing Chao](#), Jiangwen Sun, An-Li Wang, Dannie D. Langleben and Jinbo Bi, "Multi-view cluster analysis with incomplete data to understand treatment effects". Information Sciences (IF=6.9, 494, pages 278-293, 2019). DOI: 10.1016/j.ins.2019.04.039.
 20. [Guoqing Chao](#), "Discriminative K-means Laplacian Clustering". Neural Processing Letters (IF:3.08, 49, pages 393-405, 2018)
 21. [Guoqing Chao](#) and Shiliang Sun, "Consensus and complementarity Based maximum entropy discrimination for multi-view classification", Information Sciences (IF=6.9, pages 296-310, 2016).
 22. [Guoqing Chao](#) and Shiliang Sun, "Alternative multi-view maximum entropy discrimination", IEEE Transactions on Neural Networks and Learning Systems (TNNLS, IF=11.683, 27(7), pages 1445-1456, 2016).
 23. [Guoqing Chao](#) and Shiliang Sun, "Multi-kernel maximum entropy discrimination for multi-view learning", Intelligent Data Analysis (IDA, IF=0.89, 20(3), pages 481-493, 2016).

Selected Conference Publications

24. Yanxi Liu, Fangxi Liu, Yipin Hu, Yanwei Yu, Lei Meng, Yongyong Chen, [Guoqing Chao*](#). "Dual-stage Contrastive Learning-enhanced Multi-view Variational Clustering" , ICML, 2026.

25. Yipin Hu, Yanxi Liu, Fangxi Liu, Yanwei Yu, Lei Meng, Jie Wen, **Guoqing Chao***. “A Consensus Anchor-guided Hypergraph Framework For Incomplete Multi-view Clustering” , ICML, 2026.
26. Huijie Ma, Haoyuan Xin, Lei Meng, Guanzhou Ke, Yongyong Chen, **Guoqing Chao***. "Reliable Neighborhood-Aware Multi-View Outlier Detection", ICML, 2026.
27. Zhenghao Zhang, ..., **Guoqing Chao***, "Dynamic Deep Graph Learning for Incomplete Multi-View Clustering with Masked Graph Reconstruction Loss", AAAI, 2026.

28. Yuanheng Fang, [Guoqing Chao](#), Wenqiang Lei, Shaobo Li, Dianhui Chu. "CDW-CoT: Clustered Distance-Weighted Chain-of-Thoughts Reasoning", AAAI 2025.
29. [Guoqing Chao*](#), Zhenghao Zhang, Lei Meng, Jie Wen, Dianhui Chu. Federated Incomplete Multi-view Clustering with Globally Fused Graph Guidance, ICML, 2025.
30. [Guoqing Chao](#), Kaixin Xu, Xijiong Xie, Yongyong Chen. "Global Graph Propagation with Hierarchical Information Transfer for Incomplete Contrastive Multi-view Clustering" AAAI 2025.
31. Xiangming Wang, Haijin Zeng, Jiaoyang Chen, Sheng Liu, Yongyong Chen, [Guoqing Chao](#), "OTLRM: Orthogonal Learning-based Low-Rank Metric for Multi-Dimensional Inverse Problems", AAAI 2025.
32. [Guoqing Chao](#), Yi Jiang, Dianhui Chu. "Incomplete Contrastive Multi-View Clustering with High-Confidence Guiding" AAAI, 2024.
33. Jie Wen, Shijie Deng, Waikeng Wong, [Guoqing Chao](#), Chao Huang, Lunke Fei, Yong Xu, "Diffusion-based Missing-view Generation With the Application on Incomplete Multi-view Clustering", Forty-first International Conference on Machine Learning, 2024.
34. Guanzhou Ke, Yang Yu, [Guoqing Chao](#), Xiaoli Wang, Chenyang Xu, Shengfeng He, "Disentangling Multi-view Representations Beyond Inductive Bias", Proceedings of the 31st ACM International Conference on Multimedia, 2582-2590, 2023.
35. Peisong Zhang, Zhijin Wang, [Guoqing Chao](#), Yaohui Huang, Jingwen Yan, "An Oriented Attention Model for Infectious Disease Cases Prediction", IEA/AIE 2022..
36. Yuhua Kang, [Guoqing Chao](#), Xin Hu, Zhiying Tu and Dianhui Chu, "Deep Learning for Fine-Grained Image Recognition: A Comprehensive Study", Proceedings of the 2022 International Conference Computer Vision and Computational Intelligence, 31-39, 2022.
37. Bolin Zhang, Zhiying Tu, Yangqin Jiang, Shufan He, [Guoqing Chao](#), Dianhui Chu and Xiaofei He, "DGPF: A Dialogue Goal Planning Framework for Cognitive Service Conversation Bot", Proceedings of the 2021 IEEE International Conference on Web Services, 335-340, 2021.
38. [Guoqing Chao](#), Chengsheng Mao, Fei Wang, Yuan Zhao, Yuan Luo, "Supervised NMF to predict ICU mortality risk. IEEE International Conference on Bioinformatics and Biomedicine (BIBM), pages1189-1194, 2018.

39. Shiliang Sun and [Guoqing Chao](#), “Multi-view maximum entropy discrimination”, In: Proceedings of the 23rd International Joint Conferences on Artificial Intelligence (IJCAI), pages 1706-1712, 2013.
40. [Guoqing Chao](#) and Shiliang Sun, “Semi-supervised multitask learning via self-training and maximum entropy discrimination”, In: Proceedings of the 19th International Conference on Neural Information Processing (ICONIP), pages 340-347, 2012.
41. [Guoqing Chao](#) and Shiliang Sun, “Applying a multitask feature sparsity method for the classification of relations between nominal”, In: Proceedings of the 11th International Conference on Machine Learning and Cybernetics (ICMLC), pages 72-76, 2012.
42. [Guoqing Chao](#), Yuan Luo, Weiping Ding “Recent advances in supervised dimension reduction: A survey”, Machine Learning and Knowledge Extraction (Pages 341-358, 2019).

Selected Honors and Awards

2025: [Stanford's World's Top 2% Researcher](#)

2022: IEA/AIE Best Special Session Paper Award

2019: The Second Prize of Natural Science of Shanghai State ([ranked 3rd](#))

Shiliang Sun, Changshui Zhang, [Guoqing Chao](#), Xijiong Xie and Qingjiu Zhang

Editorial Service

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Applied Intelligence

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Neural Processing Letters

Leading Guest Editor,

Machine Learning and Knowledge Extraction

Editor Member Board,

Machine Learning and Knowledge Extraction

Conference Services

PC member of International Conference on Learning Representations, 2026-

PC member of Annual Conference on Neural Information Processing Systems (NeurIPS), 2025-

PC member of SIGKDD Conference on Knowledge Discovery and Data Mining, 2025-

PC member of Annual Conference on Neural Information Processing Systems, 2025-

PC member of The International Conference on Machine Learning (ICML), 2025-

PC member of ACM Multimedia, 2024-

PC member of AAAI Conference on Artificial Intelligence, 2019-

PC member of International Joint Conference on Artificial Intelligence (IJCAI), 2022-

PC member of IEEE International Conference on Data Mining (ICDM), 2022
Workshop Organizer of BIBM 2016 and 2017 workshop Machine Learning and Big Data Research for Disease Classification and Phenotyping
Workshop Organizer of ICDM 2022 workshop “Multi-View Representation Learning”

Journal Reviewing Services

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
Journal of Machine Learning Research (JMLR)
ACM Computing Surveys (CSUR)
IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
IEEE Transactions on Image Processing (TIP)
IEEE Transactions on Knowledge and Data Engineering (TKDE)
IEEE Transactions on Fuzzy Systems (TFS),
IEEE Transactions on Cybernetics
IEEE Transactions on Multimedia (TMM)
IEEE Transactions on Services Computing (TSC)

IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)

IEEE Transactions on Emerging Topics in Computational Intelligence (TETCI)
IEEE Transactions on Artificial Intelligence (TAI)
IEEE Transactions on Big Data (TBD)
ACM Transactions on Knowledge Discovery from Data (TKDD),
ACM Transactions on Intelligent Systems and Technology (ACM TIST)
Information Fusion
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